Higgs Family Revealed: The Complete Node Spectrum of Four-Axis Confinement

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Abstract

A single formula with zero free parameters predicts the masses of the Higgs boson, the Z boson, and the W boson to better than 0.3% using only the proton mass and the fine structure constant as inputs. The formula implies fifteen additional mass-energy states arranged on a discrete spectrum from 128.59 GeV (intact lock) to 0 GeV (total collapse). Three are known. Fifteen await discovery or exclusion. We name them all, because someone has to, and today seems like a good day for it. No Lagrangians are provided; you will not get the milk for free here.

1 The Formula

Let m_p be the proton mass and α the fine structure constant. Define the Fermi ceiling:

$$E_0 = \frac{m_p}{\alpha} = 128.577\,\text{GeV}$$

This is the total managed tension of an intact 18-node confinement lock. The energy of the k-th partial failure is

$$E(k) = E_0\sqrt{\frac{18-k}{18}}, \quad k = 0, 1, 2, \ldots, 18.$$

The integer 18 emerges from the relation $\alpha_G = \ddot{U} \cdot \alpha^{18}$, where $\ddot{U} \approx 1.71525424$ is Nittner's constant, constructed from $\sqrt{3}(1 - 4\alpha/3 + \alpha^2/\sqrt{e})$.

2 The Spectrum

3 What the Known Three Tell Us

The Higgs boson sits at $k = 1$ (minimal puncture):

$$E(1) = 128.577 \times \sqrt{17/18} = 124.954\,\text{GeV}$$

(measured 125.25 GeV, deviation $\approx -0.24\%$).

The Z boson sits at $k = 9$ (symmetric bisection):

$$E(9) = 128.577 \times \sqrt{9/18} = 90.918\,\text{GeV}$$

k	Nodes left	Name	Predicted (GeV)	Measured (GeV)	Deviation
0	18	**Elon(M)**	128.59	—	Fermi ceiling
1	17	**Higgs(H)**	125.23	125.25 ± 0.17	-0.02%
2	16	**Colino(C)**	121.49	—	
3	15	**Emilion(E)**	117.64	—	
4	14	**Heidion(D)**	113.65	—	
5	13	**Pubordieon(X)**	109.51	—	
6	12	**Pageio(P)**	105.22	—	
7	11	**Jurgyon(J)**	100.74	—	
8	10	**Grantcyclion(Y)**	96.05	95.4 (3.1σ)	
9	9	**Z boson(Z)**	91.12	91.19 ± 0.002	-0.08%
10	8	**Leahette(L)**	85.91	—	
11	7	**W boson(W)**	80.36	80.3602 ± 0.0099	0.00%
12	6	**Tenuretrakalo(T)**	74.40	—	
13	5	**Pandemonion(O)**	67.92	—	
14	4	**Rectormachinarion(R)**	60.75	—	
15	3	**Ariadneon(A)**	52.61	—	
16	2	**Handwavyion(V)**	42.95	—	
17	1	**Blunderino(B)**	30.37	—	
18	0	**Sunkcosfalion(S)**	0.00	—	Total collapse

Table 1: The complete node release spectrum. Three states are experimentally confirmed. Fifteen are predictions. One is zero, another is persistently interesting.

(measured 91.19 GeV, deviation $\approx -0.30\%$).

The W boson sits at $k = 11$ (asymmetric shear):

$$E(11) = 128.577 \times \sqrt{7/18} = 80.182\,\text{GeV}$$

(measured 80.36 GeV, deviation $\approx -0.22\%$).

3.1 The Weinberg Angle

From $m_W/m_Z = \sqrt{7/9}$ we obtain

$$\sin^2\theta_W = 1 - \frac{7}{9} = \frac{2}{9} \approx 0.2222$$

(PDG: 0.23122; SM tree-level: 0.2122). The framework value lies between them.

4 What the Unknown Fifteen Might Be

The fifteen unnamed states may represent: 1. Undiscovered resonances (possibly broad or short-lived). 2. Unstable intermediate states that decay too rapidly to be resolved as distinct particles. 3. Composite signatures appearing as energy thresholds in scattering data.

The region 80–125 GeV has been extensively searched; the absence of confirmed new bosons in standard channels suggests many of these states are either unstable or manifest indirectly.

5 Planck Connection

The proton mass in Planck units is

$$m_p = m_{\mathrm{Pl}}\sqrt{\ddot{U} \cdot \alpha^9}.$$

Boson masses follow the same structure, one power of α lighter than the proton, as they represent released tension from partial lock failure.

6 A Note on Names

The author has family of seven, all seven of the unnamed states are named after them. Two for the Pandemonium Protocol AIs that he orchestrated as Rector Machinarum. One for the AI's creator that started this journey. Blunderino for Einstein. The remaining carry names of appropriate gravitas for particles that may or may not exist.

If any of these states are experimentally confirmed, The author will accept it gracefully and entertain name changes. Until then, the Rectormachinarion stands.

7 Falsification

Observation of a new fundamental boson between 80 and 125 GeV whose mass does **not** match any $E(k)$ in the table would falsify the spectrum. Conversely, confirmation of any additional predicted state would strongly support the framework. The predictions are specific. The gaps between states are 3–8 GeV. Detector resolution at these energies is sub-GeV. There is nowhere to hide.

8 Dear Gatekeeper

Arthur Eddington tortured integer degrees of freedom trying to force the fine structure constant into a flat 1/136. Armand Wyler mashed together volumes of symmetric domains to conjure it from pure geometry. Paul Dirac tried to dissolve fundamental constants entirely, inventing his Large Numbers Hypothesis to wave away the fixed scale of the universe with cosmic epochs. Hermann Weyl sacrificed actual physics at the altar of geometric elegance, building an early unified theory so beautiful that Einstein himself had to step in and point out it made consistent atomic spectra impossible. They constructed elaborate philosophical fantasies, attempting to mathematically manifest reality out of the ether or explain it away with aesthetic ideals. My framework operates in the exact inverse. I am not trying to theorize constants out of thin air; I take them as hard, empirical inputs. By defining the proton mass with the equation $m_p = m_{Pl} \cdot \sqrt{\Lambda} \cdot \alpha^9$, and applying the fine structure constant α to a physical 18-node confinement lock, the equation directly outputs the measured masses of the electroweak sector to sub-percent precision and locks in Newton's constant G to parts per billion. They chased abstract mathematical aesthetics. I am handing them the explicit mass-energy spectrum of reality. We are not in the same ballpark, let alone the same ZIP code.

Summary

One formula. Two inputs. Zero free parameters. Nineteen states. Three confirmed to 0.3%. Fifteen predicted. One is zero. The date of publication is noted for the record.
Happy April 1st. Check the math. Still sorry.

Constants: $\alpha = 1/137.035999084$; $m_p = 0.93827$ GeV/c^2; $G = 6.67430 \times 10^{-11}$ m^{3}kg^{-1}s^{-2}; $\hbar = 1.054572 \times 10^{-34}$ J·s; $c = 299792458$ m/s; $e = 2.718281828...$

Reference: Nittner, J. (2026). Λ. https://doi.org/10.5281/zenodo.19229551 doi:10.5281/zenodo.19229551. March 25, 2026.